VIRGINIA COMMONWEALTH UNIVERSITY

Statistical Analysis and Modelling (SCMA 632)

A6: Visualization – Perceptual Mapping for Business

UJJWAL AKSHITH MONDRETI

V01153239

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Introduction

This analysis focuses on understanding regional disparities in food consumption in the state of **Meghalaya**, India. The study uses unit-level consumption data from the **68th round of the National Sample Survey (NSS)** conducted in 2011–12. Using both R and Python programming environments, the project visualizes and compares the average food consumption across various districts within Meghalaya.

Objectives

- To filter and analyze household-level food consumption data specifically for the state of Meghalaya.
- To compute and compare the average food consumption across districts in Meghalaya.
- To **visualize the regional patterns** using bar plots and choropleth maps in both R and Python.
- To integrate district-level data with geospatial information for map-based analysis.
- To identify any stark **disparities in consumption** levels across regions.

Business Significance

Understanding household food consumption at the regional level is crucial for:

- **Policy formulation** related to food security and poverty alleviation.
- **Targeting government schemes** like the Public Distribution System (PDS) more effectively.
- Identifying **disparities among districts**, which can support local-level development planning.
- Assessing whether food assistance is aligned with **local consumption behavior** and need.

Results

- Python
  - Computed the average *foodtotal_v* across districts.
  - Bar plot showed **South Garo Hills** and **West Garo Hills** as having the highest average food consumption.
  - Merged district-level averages with a GeoJSON file to plot a colored choropleth map of Meghalaya.
- R
  - Used *dplyr* and *ggplot2* for data wrangling and visualization.
  - Created histograms and bar plots showing food consumption patterns across districts.
  - Mapped average food consumption using *sf* package and the same GeoJSON file.
  - Clearly showed **higher consumption in Garo Hills Regions**, similar to Python analysis.

Interpretations

- **South Garo Hills** consistently exhibited the **highest average food consumption**, suggesting a higher standard of living or dietary expenditure.
- **East and West Khasi Hills** were mid-range in terms of food spending.
- Several districts like **East Jaintia Hills** and newly created ones showed **zero or missing values**, likely due to incomplete or misaligned mapping data.
- Both platforms confirmed similar trends, providing **cross-validation of results**.

Recommendations

- Data Enrichment: Future datasets should ensure proper alignment between district codes and district names to avoid mismatches and missing data in visualizations.
- Targeted Policy: Policymakers should examine why certain districts like South Garo Hills have such high consumption and whether it correlated with income or other demographic indicators.
- Infrastructure for Data Collection: More granular and recent data collection can improve accuracy in identifying current trends.
- Interactive Dashboards: For practical use in government and research settings, these static visualizations can be enhanced using tools like Shiny (R) or Plotly Dash (Python).

References

- Data provided through Canvas Portal.
- Meghalaya GeoJSON downloaded from this GitHub Repository.